

COMPLETE COMPUTER DESCRIPTION

Microoperations

Fetch	$R'T_0:$	$AR \leftarrow PC$
	$R'T_1:$	$IR \leftarrow M[AR], PC \leftarrow PC + 1$
Decode	$R'T_2:$	$D_0, \dots, D_7 \leftarrow \text{Decode } IR(12 \sim 14), AR \leftarrow IR(0 \sim 11), I \leftarrow IR(15)$
Indirect	$D_7'IT_3:$	$AR \leftarrow M[AR]$
Interrupt	$T_0'T_1'T_2'(IEN)(FGI + FGO):$	$R \leftarrow 1$
	$RT_0:$	$AR \leftarrow 0, TR \leftarrow PC$
	$RT_1:$	$M[AR] \leftarrow TR, PC \leftarrow 0$
	$RT_2:$	$PC \leftarrow PC + 1, IEN \leftarrow 0, R \leftarrow 0, SC \leftarrow 0$
Memory-Reference		
AND	$D_0T_4:$	$DR \leftarrow M[AR]$
	$D_0T_5:$	$AC \leftarrow AC \wedge DR, SC \leftarrow 0$
ADD	$D_1T_4:$	$DR \leftarrow M[AR]$
	$D_1T_5:$	$AC \leftarrow AC + DR, E \leftarrow C_{out}, SC \leftarrow 0$
LDA	$D_2T_4:$	$DR \leftarrow M[AR]$
	$D_2T_5:$	$AC \leftarrow DR, SC \leftarrow 0$
STA	$D_3T_4:$	$M[AR] \leftarrow AC, SC \leftarrow 0$
BUN	$D_4T_4:$	$PC \leftarrow AR, SC \leftarrow 0$
		$PC \leftarrow AR, SC \leftarrow 0$
BSA	$D_5T_4:$	$M[AR] \leftarrow PC, AR \leftarrow AR + 1$
	$D_5T_5:$	$PC \leftarrow AR, SC \leftarrow 0$
ISZ	$D_6T_4:$	$DR \leftarrow M[AR]$
	$D_6T_5:$	$DR \leftarrow DR + 1$
	$D_6T_6:$	$M[AR] \leftarrow DR, \text{ if } (DR=0) \text{ then } (PC \leftarrow PC + 1), SC \leftarrow 0$



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Register-Reference

	$D_7I'T_3 = r$	(Common to all register-reference instruction)
	$IR(i) = B_i$	($i = 0,1,2, \dots, 11$)
	$r:$	$SC \leftarrow 0$
CLA	$rB_{11}:$	$AC \leftarrow 0$
CLE	$rB_{10}:$	$E \leftarrow 0$
CMA	$rB_9:$	$AC \leftarrow AC'$
CME	$rB_8:$	$E \leftarrow E'$
CIR	$rB_7:$	$AC \leftarrow shr\ AC, AC(15) \leftarrow E, E \leftarrow AC(0)$
CIL	$rB_6:$	$AC \leftarrow shl\ AC, AC(0) \leftarrow E, E \leftarrow AC(15)$
INC	$rB_5:$	$AC \leftarrow AC + 1$
SPA	$rB_4:$	If($AC(15) = 0$) then ($PC \leftarrow PC + 1$)
SNA	$rB_3:$	If($AC(15) = 1$) then ($PC \leftarrow PC + 1$)
SZA	$rB_2:$	If($AC = 0$) then ($PC \leftarrow PC + 1$)
SZE	$rB_1:$	If($E = 0$) then ($PC \leftarrow PC + 1$)
HLT	$rB_0:$	$S \leftarrow 0$

Input-Output

	$D_7IT_3 = p$	(Common to all input-output instructions)
	$IR(i) = B_i$	($i = 6,7,8,9,10,11$)
	$p:$	$SC \leftarrow 0$
INP	$pB_{11}:$	$AC(0-7) \leftarrow INPR, FGI \leftarrow 0$
OUT	$pB_{10}:$	$OUTR \leftarrow AC(0-7), FGO \leftarrow 0$
SKI	$pB_9:$	If($FGI = 1$) then ($PC \leftarrow PC + 1$)
SKO	$pB_8:$	If($FGO = 1$) then ($PC \leftarrow PC + 1$)
ION	$pB_7:$	$IEN \leftarrow 1$
IOF	$pB_6:$	$IEN \leftarrow 0$



DESIGN OF BASIC COMPUTER(BC)

- Hardware Components of BC
 - A memory unit: 4096 x 16.
 - Registers:
AR, PC, DR, AC, IR, TR, OUTR, INPR, and SC
 - Flip-Flops(Status):
I, S, E, R, IEN, FGI, and FGO
 - Decoders: 3x8 Opcode decoder
 - 4x16 timing decoder
 - Common bus: 16 bits
 - Control logic gates:
 - Adder and Logic circuit: Connected to the input of AC
- Control Logic Gates (Output)
 - Signals to control the inputs of the nine registers
 - Signals to control the Read and Write inputs of memory
 - Signals to control Set, Clear, or Complement the flip-flops
 - Signals for S_2 , S_1 , S_0 to select a register for the bus
 - Signals to control the AC Adder and Logic circuit



CONTROL OF REGISTERS AND MEMORY

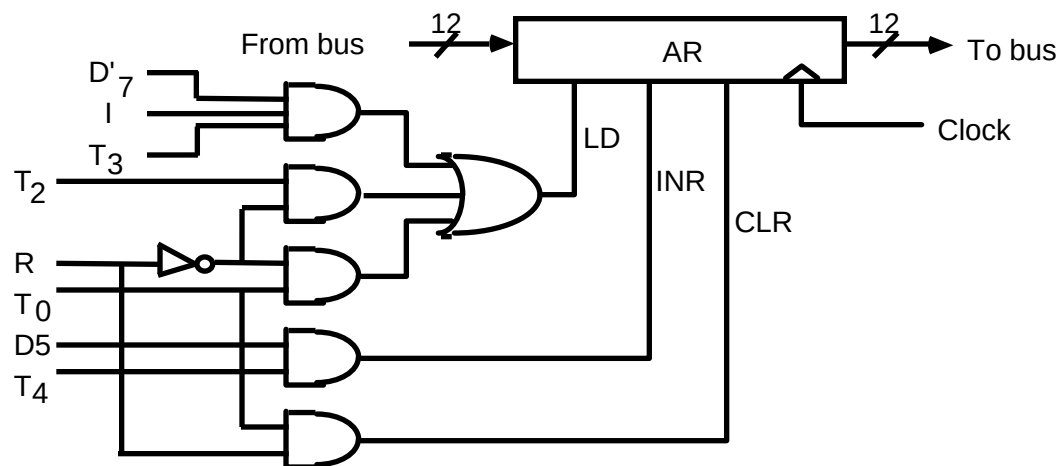
Address Register: AR

Scan all of the register transfer statements that change the content of AR:

$R'T_0$:	$AR \leftarrow PC$	$LD(AR)$
$R'T_2$:	$AR \leftarrow IR(0-11)$	$LD(AR)$
D'_7IT_3 :	$AR \leftarrow M[AR]$	$LD(AR)$
RT_0 :	$AR \leftarrow 0$	$CLR(AR)$
D_5T_4 :	$AR \leftarrow AR + 1$	$INR(AR)$



$LD(AR) = R'T_0 + R'T_2 + D'_7IT_3$
$CLR(AR) = RT_0$
$INR(AR) = D_5T_4$



CONTROL OF FLAGS

IEN: Interrupt Enable Flag

pB_7 : $IEN \leftarrow 1$ (I/O Instruction)

pB_6 : $IEN \leftarrow 0$ (I/O Instruction)

RT_2 : $IEN \leftarrow 0$ (Interrupt)

$p = D_7IT_3$ (Input/Output Instruction)

